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INTRODUCTION

- Peripheral giant cell granuloma (PGCG) is a **benign reactive lesion** occurring exclusively on the gingiva or edentulous alveolar ridge, often associated with chronic local irritation.
- Typically presents as a red to purplish nodular mass and may rarely cause underlying bone erosion.
- Histologically, PGCG is characterized by multinucleated giant cells within a vascular stroma. Given its histological overlap with other giant cell-rich lesions, clinical, radiological and biochemical information from the treating team is essential for accurate diagnosis.



"ACCURATE DIAGNOSIS IS A TEAM EFFORT—CLINICAL CONTEXT GUIDES THE PATHOLOGIST, WHILE HISTOPATHOLOGY GUIDES THE CLINICIAN"

CASE PRESENTATION



65 year-old

Chief complaint:

Asymptomatic growth on the left posterior mandibular alveolar ridge since July 2025.

History of chief complaint:

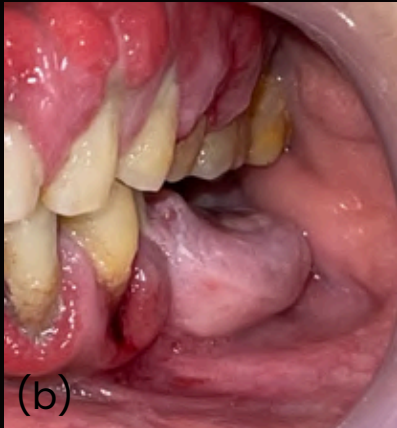
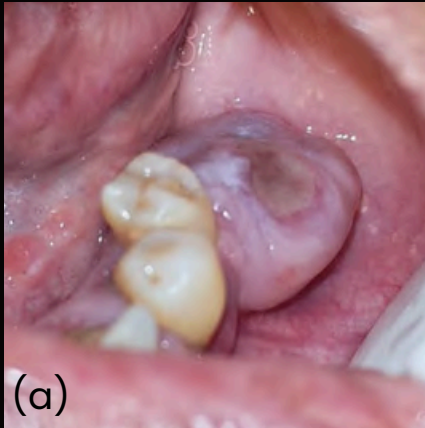
Incidental finding of swelling over left lower posterior ridge when he went for extraction of mobile lower front teeth.

Medical history:

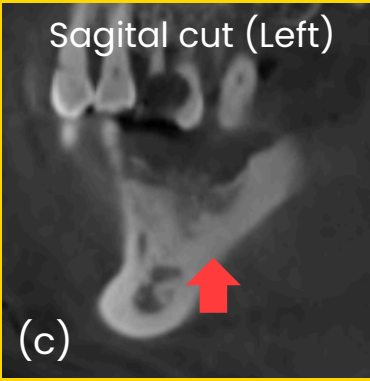
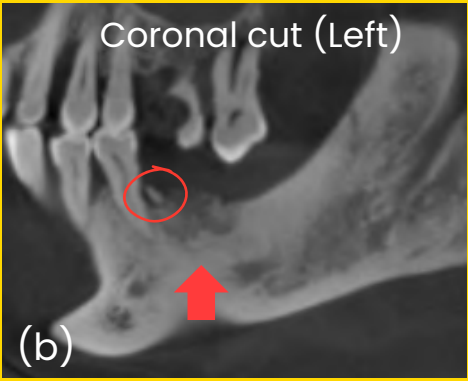
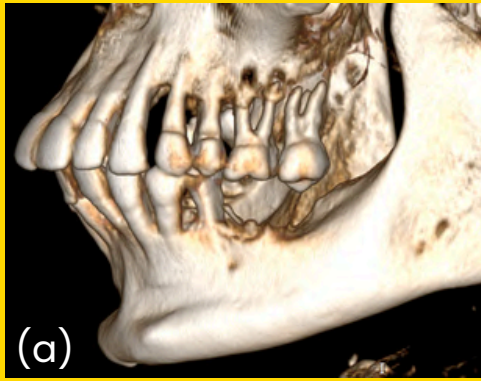
- 01 Diabetes mellitus
- 02 Hypertension
- 03 Dyslipidaemia
- 04 Chronic kidney disease stage 3b

Clinical examination

- Extra-oral findings**
 - Facial symmetry.
 - No lymph nodes palpable.
- Intra-oral findings**
 - A bony hard pedunculated mass measuring 3.0 x 1.5 cm (Figures a & b).
 - Located in the left lower edentulous region corresponding to teeth 36–38.
 - Overlying normal mucosa colour, no pus discharge or bleeding.
 - Non-tender on palpation.



Radiographic findings

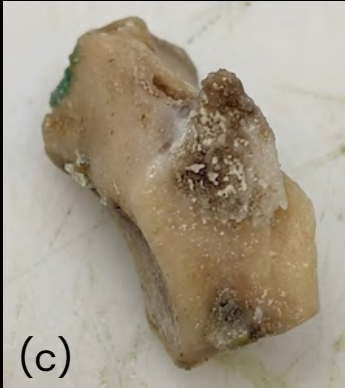
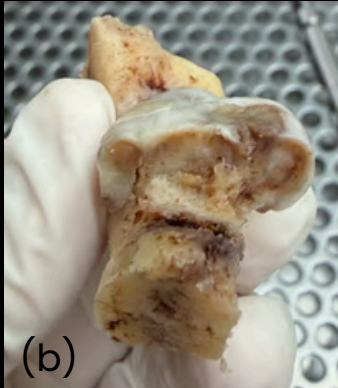


- Cone beam computed tomography (CBCT) (Figures a-c) demonstrated superficial saucerisation of the superior alveolar ridge with a slight mixed radiolucent-radiopaque appearance and a retained root (circle).
- No buccolingual cortical expansion.

Biochemical findings

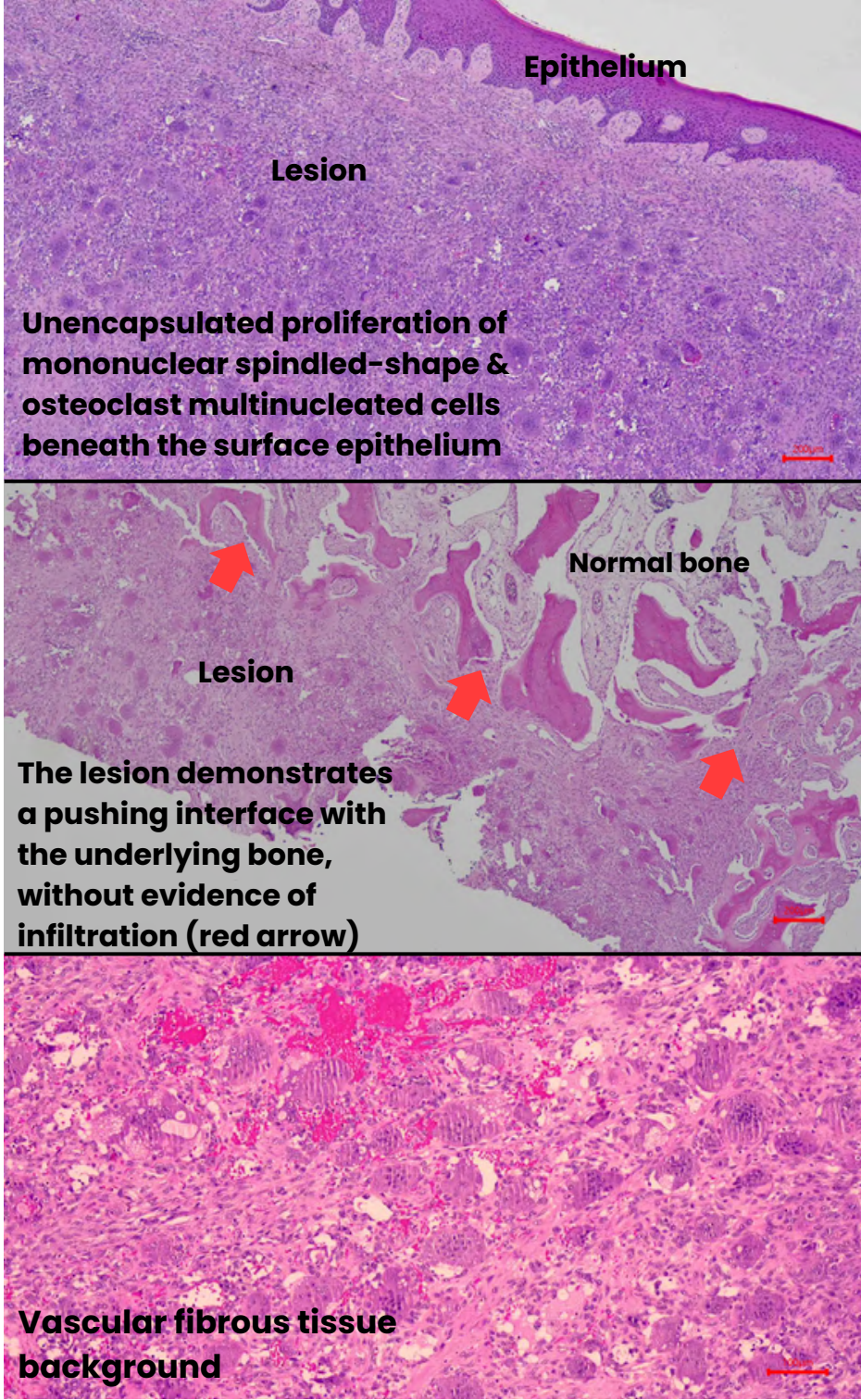
Serum investigation	Result	Reference range
Calcium	2.26 mmol/L	2.20-2.65 mmol/L
Parathyroid hormone (PTH)	8.60 pmol/L	1.3-9.30 pmol/L
Alkaline phosphatase (ALP)	80 U/L	43-115 U/L

Macroscopic findings



- A marginal resection of the left posterior mandible, comprising a pedunculated, firm, nodular mass overlying the alveolar ridge (Figure a).
- The mass has a smooth surface and measures 21 (AP) × 20 (BL) × 16 (SI) mm.
- The cut surface is solid, lobulated and tan-coloured, with firm adherence to the underlying mandible (Figures b and c).
- A retained root is identified (not shown).

Photomicrographs



Unencapsulated proliferation of mononuclear spindled-shape & osteoclast multinucleated cells beneath the surface epithelium

The lesion demonstrates a pushing interface with the underlying bone, without evidence of infiltration (red arrow)

Discussion

- The **main differential diagnoses** for a giant cell-rich lesion over the jaw:
 - Peripheral giant cell granuloma (PGCG)
 - Central giant cell granuloma (CGCG)
 - Brown tumour
- The peripheral location, superficial bone involvement and pushing rather than infiltrative growth pattern favour **PGCG** over **CGCG**, which is primarily an intraosseous lesion.
- The presence of an retained root may have provided a source of chronic local irritation, supporting the reactive nature of PGCG.
- Brown tumour** was considered given the patient's chronic kidney disease. However, the peripheral location and normal serum calcium, PTH and ALP levels provide no radiological or biochemical support for brown tumour.
- Although **KRAS mutations** have been reported in both **PGCG** and **CGCG**, this molecular overlap does not reliably distinguish between the two entities, highlighting the importance of clinicoradiological and histopathological correlation.

CASE UPDATE

At 7 months postoperatively, no recurrence was observed.

REFERENCES

WHO Classification of Head and Neck Tumours (5th edn).

ACKNOWLEDGEMENT

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DECLARATION OF CONFLICT FOR ALL AUTHORS

The authors declare no conflict of interest.

Conclusion

- Clinicoradiological, biochemical and histopathological correlation is essential for accurate diagnosis of PGCG.
- PGCG may recur following excision, with recurrence rates of up to 16% reported. In this case, marginal resection was performed, with removal of the potential source of local irritation.